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CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 99-077

WASTE DISCHARGE REQUIREMENTS
FOR
HILMAR COUNTY WATER DISTRICT
WASTEWATER TREATMENT FACILITY
MERCED COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. The Hilmar County Water District (hereafter Discharger) submitted a Report of Waste Discharge (RWD) including a technical report with process design information and a water balance dated 14 April 1998, a Mitigated Negative Declaration/Initial Study dated 27 February 1998, and supplemental information in support of the replacement and relocation of its existing wastewater treatment facility (WWTF) and an increase in discharge flow. The Discharger's wastewater is from residences and commercial establishments. The property (Assessor's Parcel Nos. 45-210-39 and 45-220-30) is owned by the Discharger.
2. Waste Discharge Requirements Order No. 85-228, adopted by the Board on 23 August 1985, prescribes requirements for a monthly average design discharge of 0.30 million gallons per day (mgd) of wastewater from an aeration pond to two evaporation/percolation ponds at another location (Assessor's Parcel No. 45-21-55).
3. The monthly self-monitoring reports submitted by the Discharger from 1994 to present indicate that monthly dry weather flows have averaged 97% of the WWTF's rated treatment capacity.
4. On 20 June 1997, the Board adopted Cease and Desist (C&D) Order No. 97-103 due to the lack of WWTF capacity and potential nuisance odor conditions from low dissolved oxygen concentrations in the treatment and evaporation/percolation ponds. In addition, C&D Order No. 97-103 found the existing WWTF degraded local groundwater with total coliform, nitrate, BOD, and COD, and with salts as evidenced by conductivity at 25°C (EC) and total dissolved solids (TDS).
5. The Discharger proposes to replace the existing WWTF with an Advanced Integrated Wastewater Pond SystemTM (hereafter referred to as WWTF) and discharge a maximum monthly average daily flow of 0.55 mgd of treated wastewater to 17.5 acres of percolation ponds. The WWTF will consist of headworks, two advanced facultative ponds (AFP), four fermentation pits, two high rate ponds, two algae settling ponds, a maturation pond, and four percolation ponds. Two to three aerators will be installed in each of the AFPs to provide supplemental aeration during start-up. The slopes and bottoms of all the treatment ponds, except for the maturation pond, will be lined with a high density

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polyethylene (HDPE), geosynthetic clay liner (GCL), shotcrete, or combination thereof to minimize infiltration to groundwater. Only the slopes of the maturation pond will be lined. The treatment ponds (except for the maturation pond) have been designed to treat 1.0 mgd of wastewater. Effluent from the maturation pond will be discharged initially to 17.5 acres of new percolation ponds. The WWTF flow diagram and layout are shown in Attachments A and B, respectively. The Discharger submitted a water balance that shows the initial 17.5 acres of percolation ponds will be adequate to dispose of an average daily flow of 0.55 mgd. The Discharger will acquire additional ground in the future to enlarge the maturation pond and percolation area, at which time it will submit a Report of Waste Discharge (RWD) to increase the allowable flow to 1.0 mgd.

6. The WWTF will be in Sections 24 & 25, T6S, R10E, MDB&M. The WWTF site is less than a mile from the Merced River and the concrete-lined Highline Canal flows through the southwestern corner of the site, as shown in Attachment C, which is attached hereto and part of this Order by reference. The site lies within the San Joaquin Valley Floor Hydrologic Unit, Turlock Hydrologic Area (No. 535.50), as depicted on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.
7. The RWD indicates that the WWTF influent and the maturation pond effluent will exhibit the following characteristics:

<u>Constituent</u>	<u>Units</u>	<u>Influent</u>	<u>Effluent</u>
BOD ¹	mg/l	200	40
Total Suspended Solids	mg/l	240	40
Total Dissolved Solids	mg/l	620	750*
EC	µmhos/cm	1100	1300*
COD	mg/l	570	140
pH	units	7.7	7.5
Total Nitrogen as N	mg/l	31	10

¹ Five day, 20°C BOD test.

* Estimated effluent values higher than influent due to concentration from evaporation.

8. The existing Hilmar Water District WWTF (Assessor's Parcel No. 45-21-55) will be leveled and the property held in reserve to meet future needs of the District.
9. Order No. 85-228 is inadequate because it does not reflect the proposed volume of discharge flow or describe the WWTF or site now proposed by the Discharger.
10. The Board adopted a Water Quality Control Plan, Fourth Edition, for the Sacramento River Basin and the San Joaquin River Basin (hereafter Basin Plan), which designates beneficial uses, establishes water quality objectives, and contains implementation plans and policies for all waters of the basins. These requirements implement the Basin Plan.

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11. The beneficial uses of the Merced River from McSwain Reservoir to the San Joaquin River are municipal and domestic supply, agricultural supply, industrial service supply, industrial process supply, hydropower generation, water contact recreation, non-contact water recreation, warm and cold freshwater habitat, migration of aquatic organisms; spawning, reproduction, and/or early development; and wildlife habitat.
12. The beneficial uses of underlying groundwater are municipal and domestic supply, agricultural supply, industrial service supply, and industrial process supply.
13. The Basin Plan encourages reclamation. The Discharger reviewed opportunities for reclamation and, due to other readily available and relatively inexpensive sources of water, found use of effluent on irrigated land economically infeasible.
14. The RWD indicates that the soils underlying the WWTF are silty sands to depths of between 7.5 and 13 feet below ground surface (bgs). The silty sands are underlain by interbedded strata of sandy silts, poorly graded sands, and more silty sands to the maximum depth explored in this area of 41.5 feet bgs. Permeability rates are expected to be moderate to moderately rapid in character. The disposal area has been designed using a 0.17 ft/day percolation rate.
15. The technical report indicates that groundwater beneath the WWTF's 60-acre parcel ranges in depth from 7.5 feet bgs in the north portion to 27 feet bgs in the south west portion. Depth to groundwater beneath the proposed percolation bed area is approximately 16.5 feet bgs. Groundwater flow is to the south/southwest. The WWTF has five monitoring wells (MW1 through MW5) near the property's four boundaries. Groundwater data was collected once for each well during January and February of 1998 and again during June 1998. Groundwater samples taken from MW1 through MW5 indicate TDS concentrations of between 140 mg/l and 440 mg/l. Total coliform levels from MW1 through MW5 have measured greater than 23 MPN/100ml. Monitoring well samples indicate nitrate concentrations of between 17 mg/l and 170 mg/l. Agricultural activities upgradient of the WWTF site probably account for these contaminant levels.
16. The WWTF site is bordered on the north by Geer Avenue, on the east by Griffith Avenue, and on the south and west by parallel running Highline Canal / Williams Avenue and agricultural land.
17. The approximate annual average precipitation at Hilmar is 12.2 inches and the annual average pan evaporation is approximately 61 inches.

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18. The water balance is based on a total annual precipitation with a 100-year return period. The total annual 100-year rainfall is reported as 26.16 inches. The water balance does not indicate significant infiltration or inflow.
19. The Hilmar County Water District adopted a Resolution Issuing a Mitigated Negative Declaration on 5 May 1998 for the proposed project in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) and State CEQA Guidelines. The Board has reviewed the negative declaration and concurs that the project as approved will not have a significant effect on water quality. Mitigation measures to protect against the degradation of groundwater include the placement of monitoring wells around the perimeter of the site to measure groundwater quality and elevation.
20. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
21. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that the Hilmar County Water District, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following at the above described WWTF:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Discharge of waste classified as 'hazardous,' as defined in Section 2521(a) of Title 23, California Code of Regulations (CCR), Section 2510, et seq., (hereinafter Chapter 15), or 'designated,' as defined in Section 13173 of the California Water Code, is prohibited.

B. Discharge Specifications:

1. The monthly average daily discharge flow shall not exceed 0.55 million gallons.

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2. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.
3. As a means of discerning compliance with Discharge Specification No. B.2, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.
4. The treatment facility shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
5. The effluent from the WWTF shall not exceed the following limits:

<u>Constituent</u>	<u>Units</u>	<u>Monthly Average</u>	<u>Daily Maximum</u>
BOD ₅ ¹	mg/l	40	80
Total Suspended Solids	mg/l	40	80
Settleable Solids	ml/l	0.2	0.5

¹ 5-day, 20°C BOD test.

6. Percolation ponds shall not have a pH less than 6.5 or greater than 9.5.
7. Percolation ponds shall be managed to prevent breeding of mosquitoes. In particular:
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
8. Public contact with wastewater in the treatment and disposal areas shall be precluded through such means as fences and signs, or acceptable alternatives.
9. Percolation ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration. Design seasonal precipitation shall be based on total annual

precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet in any pond (measured vertically).

10. On or about 1 October of each year, available pond storage capacity shall be at least equal to the volume necessary to comply with Discharge Specification B.9.

C. Sludge Disposal:

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer and consistent with *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in title 27, California Code of Regulations, Division 2, Subdivision 1, Section 20005, et seq.
2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer at least 60 days in advance of the change.
3. Use and disposal of sewage sludge shall comply with existing federal, state, and local laws, regulations, and ordinances, including permitting requirements and technical standards included in 40 CFR 503 and the requirements of Merced County Sludge Ordinance #1505.

If the State Water Resources Control Board and the Regional Water Quality Control Boards assume primacy to implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

D. Groundwater Limitations:

The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentrations statistically greater than background groundwater quality. For purposes of comparison, background water quality shall be determined when groundwater monitoring provides sufficient data. Quality determined in this manner establishes "water quality protection standards." A sufficient number of upgradient and down gradient groundwater monitoring wells shall be established for the percolation ponds. The down gradient wells shall constitute "points of compliance" (POCs). In conjunction with background monitoring, monitoring of POCs will enable one to determine compliance with water quality protection standards.

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E. Provisions:

1. This Order is intended to regulate discharge from the replacement WWTF once it is operational and does not supersede Waste Discharge Requirements (WDRs) Order No. 85-228 which regulates discharge from the existing WWTF. The Discharger shall submit a written certification to the Board at least 30 days prior to directing wastewater from the existing WWTF to the replacement WWTF. The certification shall be appropriately signed and stamped by a California registered civil engineer and shall certify whether the replacement WWTF was completed as designed and is ready for operation. (Following diversion of all flows to the WWTF regulated by this Order, Order No. 85-228 shall be scheduled for rescission by the Board).
2. The Discharger shall comply with Monitoring and Reporting Program No. 99-077, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
3. The Discharger shall not allow pollutant-free wastewater to be discharged into the collection, treatment, and disposal system in amounts that significantly diminish the system's capability to comply with this Order. Pollutant-free wastewater means rainfall, groundwater, cooling waters, and condensates that without treatment are essentially free of pollutants.
4. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
5. **At least 90 days** prior to sludge or solids removal from the WWTF, the Discharger shall submit a sludge production and disposal plan describing proposed sludge management/disposal practices and demonstrating how the proposed practices will achieve compliance with Sludge Disposal Specification Nos. C.1 through C.3. The report shall be prepared and signed by a California registered civil engineer experienced in waste treatment and disposal.
6. A copy of the operation and maintenance manual for the WWTF shall be submitted **30 days prior to discharge to the WWTF**. The operation and maintenance manual shall be prepared under the supervision of a California registered civil engineer and include descriptions of complete operational procedures for the WWTF that assure compliance with this Order. Further, the operation and maintenance manual must satisfy the items in Standard Provision B.2.

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7. **By at least 120 days prior to start-up of the WWTF**, the Discharger shall submit a technical report in the form of a work plan to install and develop a comprehensive groundwater monitoring network to monitor at the earliest practicable date the effects of its disposal operations on the uppermost groundwater aquifer. The proposed network shall include, but not be limited to: (1) a sufficient number of upgradient and downgradient groundwater monitoring wells to monitor the effect of the percolation ponds.¹, (2) a map showing the proposed location of the wells, and (3) the proposed well specifications. The Discharger shall install the network within 90 days of the Executive Officer's approval of the work plan and monitor all monitoring wells as described in the attached Monitoring and Reporting Program.
8. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

To assume operation under this Order, the succeeding owner or operator must apply in writing to the Executive Officer requesting transfer of the Order. The request must contain the requesting entity's full legal name, the State of incorporation if a corporation, the name and address and telephone number of the persons responsible for contact with the Board, and a statement. The statement shall comply with the signatory paragraph of Standard Provision B.3 and state that the proposed owner or operator assumes full responsibility for compliance with this Order. Failure to submit the request shall be considered a discharge without requirements, a violation of the California Water Code. Transfer shall be approved or disapproved by the Executive Officer.

9. The Discharger shall use the best practicable control technique available to comply with this Order.
10. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer.

Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.

¹ The downgradient wells shall constitute the "points of compliance" (POCs) referenced in Section D. In conjunction with upgradient or background monitoring, monitoring of POCs will enable one to establish and then determine compliance with water quality protection standards.

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11. A copy of this Order shall be kept at the WWTF for reference by operating personnel. Key operating personnel shall be familiar with its contents.
12. If reclaimed water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes." Other uses of reclaimed water not specifically authorized herein shall be subject to revision of this Order and compliance with Title 22, CCR, Division 4.
13. The Discharger shall implement the necessary legal authorities, programs, and controls to ensure that the following incompatible wastes are not introduced to the treatment system, where incompatible wastes are:
 - a. Wastes which create a fire or explosion hazard in the treatment works;
 - b. Wastes which will cause corrosive structural damage to treatment works, but in no case wastes with a pH lower than 5.0, unless the works is specially designed to accomodate such wastes;
 - c. Solid or viscous wastes in amounts which cause obstruction to flow in sewers, or which cause other interference with proper operation or treatment works;
 - d. Any waste, including oxygen demanding pollutants (BOD, etc.) released in such volume or strength as to cause inhibition or disruption in the treatment works;
 - e. Heat in amounts that inhibit or disrupt biological activity in the treatment works, or that raise influent temperatures above 40 °C (104 °F), unless the treatment works is designed to accomodate such heat;
 - f. Petroleum oil, nonbiodegradable cutting oil, or products of mineral oil origin in amounts that will cause interference or pass through;
 - g. Pollutants which result in the presence of toxic gases, vapors, or fumes within the treatment works in a quantity that may cause acute worker health and safety problems; and
 - h. Any trucked or hauled pollutants, except at points predesignated by the discharger.
14. The Board will review this Order periodically and will revise requirements when necessary.

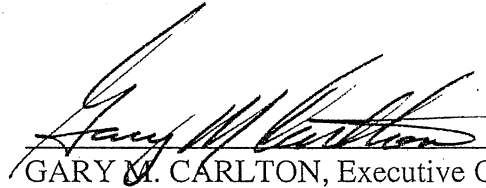
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15. The Discharger shall notify the Board **at least 120 days prior to:**

- a. The addition or any significant introduction of pollutants into the WWTF by an industrial user that might substantially change the character of the wastewater from that described in the RWD.
- b. Any planned physical alterations or additions to the WWTF, or changes planned in the discharger's sludge use or disposal practice, where such alterations, additions, or changes may justify the application of waste discharge requirements that are different from or absent in the existing waste discharge requirements.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 11 June 1999.



GARY M. CARLTON, Executive Officer

CTS:cts/fmc:6/11/99

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
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Specific sample station locations shall be established with concurrence of the Board's staff. A description of the stations shall be submitted to Board and attached to this Program.

INFLUENT MONITORING

Influent samples shall be collected at the inlet of the headworks and at approximately the same time as effluent samples. Influent monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
BOD ₅ ¹	mg/l	8-hr. Composite ²	2/Month ³
Total Suspended Solids	mg/l	8-hr. Composite ²	2/Month ³
Settleable Solids	ml/l	Grab	2/Month ³

¹ Five day, 20°C BOD

² Composite samples may consist of flow-proportioned grab samples.

³ During non-consecutive weeks.

EFFLUENT MONITORING

Except for flow, which may be measured either at the headworks or outlet from the treatment system, effluent samples shall be collected just prior to discharge to the percolation ponds. Effluent samples shall be representative of the volume and nature of the discharge. Time of collection of grab samples shall be recorded. Effluent monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Flow	mgd	Metered	Continuous ¹
EC ²	µmhos/cm	Grab	Weekly
pH	pH Units	Grab	Weekly

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<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
BOD ₅ ³	mg/l	Grab	2/Month ⁴
Total Suspended Solids	mg/l	Grab	2/Month ⁴
Settleable Solids	ml/l	Grab	2/Month ⁴
Total Kjeldahl Nitrogen	mg/l	Grab	2/Year ⁵
Nitrates as N	mg/l	Grab	2/Year ⁵
Standard Minerals ⁶	mg/l	Grab	Annually ⁷

¹ Recorded Daily.

² Conductivity @ 25°C.

³ Five day, 20°C biological oxygen demand test.

⁴ During non-consecutive weeks.

⁵ June/December.

⁶ Standard Minerals shall include all major cations and anions and include a verification that the analysis is complete (i.e., a cation/anion balance).

⁷ December.

POND MONITORING

Percolation ponds shall be monitored as follows:

<u>Constituent</u>	<u>Unit</u>	<u>Type of Sample</u>	<u>Frequency</u>
pH	pH Units	Grab	Weekly
Freeboard	feet	Observation	Weekly
Dissolved Oxygen	mg/l	Grab ¹	Weekly ²

¹ Samples shall be collected at a depth of one foot, opposite the inlet. Dissolved Oxygen samples shall be collected between 0800 and 0900 hours.

² If results indicate a concentration of less than 1.0 mg/l or if offensive odors are noted, the frequency of monitoring shall be increased as necessary to characterize the period of non-compliance.

Permanent markers shall be placed in percolation ponds with calibrations indicating the water level at design capacity and available operational freeboard. In addition, the Discharger shall

inspect the condition of the ponds once per week and write visual observations in a bound log book. Notations shall include observations of whether weeds are developing in the water or along the bank, and their location; whether burrowing animals or insects are present; and the color of the pond (e.g. dark sparkling green, dull green, yellow, gray, tan, brown, etc.). A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month. Where the Operation and Maintenance Manual indicates remedial action is necessary, the Discharger shall briefly explain in the transmittal what action has been taken or is scheduled to be taken.

GROUNDWATER MONITORING

The Discharger shall submit a work plan at least **120 days** prior to discharge to the WWTF for a proposed groundwater monitoring network, with a schedule for implementation to monitor groundwater in the area of the WWTF. The time schedule shall be such that, following Board approval, the groundwater monitoring network shall be installed and the first round of samples taken at least **30 days** prior to discharge to the WWTF. The monitoring network shall consist of one or more background monitoring wells and sufficient downgradient wells to determine flow direction and gradient, and to monitor disposal areas. All well locations and construction features are subject to the prior approval of the Executive Officer and must be sufficient to monitor potential impacts of the disposal operation on the uppermost groundwater aquifer. Existing wells proposed for inclusion in the program shall have known construction features (depth, length of perforated interval, surface seal, etc.). All monitoring wells must comply with the California Well Standards as prescribed in Department of Water Resources Bulletin Nos. 74-81 and 74-90.

Samples shall be taken monthly from approved background monitoring well(s) for one year and analyzed for the parameters specified below. Data from these analyses shall be reported to the Board **by 30 days after said year ends**, for use in determining water quality protection standards.

If subsequent sampling of the background monitoring well(s) indicates significant water quality changes due to either seasonal fluctuations or other reasons unrelated to waste disposal activities, the Discharger may request modification of the water quality protection standards.

The down gradient wells shall constitute "points of compliance" (POCs). In conjunction with background monitoring, monitoring of POCs will enable one to determine compliance with water quality protection standards. Water samples shall be collected from wells in the approved monitoring network and analyzed as follows:

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<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u>
Depth and Elevation ¹	feet	Measurement	Monthly
COD ²	mg/l	Grab	Quarterly
Conductivity	µmhos/cm	Field	Quarterly
pH	pH units	Field	Quarterly
Total Coliform Organisms	MPN ³ /100 ml	Grab	Quarterly
Standard Minerals ⁴	mg/l	Grab	Quarterly
Total Dissolved Solids ⁵	mg/l	Grab	Quarterly

¹ Elevation Relative to Mean Sea Level measured to the nearest 0.1 foot.

² Chemical oxygen demand.

³ Most Probable Number.

⁴ Standard Minerals shall include all major cations and anions including nitrate and shall include a verification that the analysis is complete (i.e., a cation/anion balance).

⁵ Determined by U.S. EPA Methods 160.1 and 160.4.

Following each sampling event, the Discharger shall determine whether there is a statistically significant increase over water quality protection standards for each parameter and constituent analyzed. If the Discharger finds there is a statistically significant increase in indicator parameters or waste constituents over the water quality protection standards at the POCs, the Discharger shall notify the Board within 30 days.

SLUDGE MONITORING

The Discharger shall monitor the sludge production and quality in accordance with the sludge production and disposal plan as approved by the Executive Officer.

WATER SUPPLY MONITORING

A sampling station shall be established where a representative sample of the water supply can be obtained. Water supply monitoring shall include at least the following:

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<u>Constituents</u>	<u>Units</u>	<u>Sampling Frequency</u>
EC	µmhos/cm	Quarterly
Total Dissolved Solids ¹	mg/l	Quarterly
Standard Minerals ²	mg/l	Quarterly

¹ Determined by EPA Methods 160.1 and 160.4

² Standard Minerals shall include all major cations and anions and include a verification that the analysis is complete (i.e., a cation/anion balance).

If the source water is from more than one source, the results shall be reported as a flow weighted average and include copies of supporting calculations.

REPORTING

Daily, weekly, 2/month, and monthly monitoring data shall be included in the monthly monitoring reports submitted to the Board by the **20th day** of the month following sample collection. Monitoring data required 2/year shall be submitted to the Board by the **20th day** of the month following the sample collection. Quarterly monitoring data shall be submitted to the Board by the **20th day** of the month following each calendar quarter. Annual monitoring data shall be submitted with the Annual Report described below.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner that illustrates clearly whether the Discharger complies with waste discharge requirements, including calculation of all averages, etc.

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to the Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss the corrective actions taken and planned to bring the discharge into full compliance with the waste discharge requirements.

By **31 January of each year**, the Discharger shall submit a written Annual Report to the Executive Officer containing the following:

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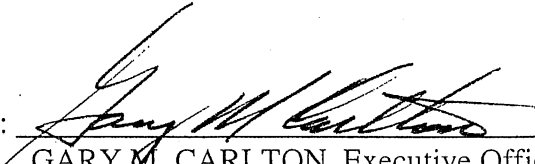
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- a. The names, titles, certificate grade, and general responsibilities of persons operating and maintaining the wastewater treatment facility.
- b. The names and telephone numbers of persons to contact regarding the plant for emergency and routine situations.
- c. A certified statement of when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who did the calibration (Standard Provision C.4).
- d. A statement whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment plant as currently constructed and operated, and the dates when these documents were last reviewed for adequacy.
- e. The total quantity of sludge disposed of during the previous year and ultimate disposal site(s).

All reports submitted in response to this Order shall comply with the signatory requirements in Standard Provision B.3.

Except for groundwater monitoring, which shall be initiated no later than specified herein, the Discharger shall implement the above monitoring program during the first week of operation of the WWTF.

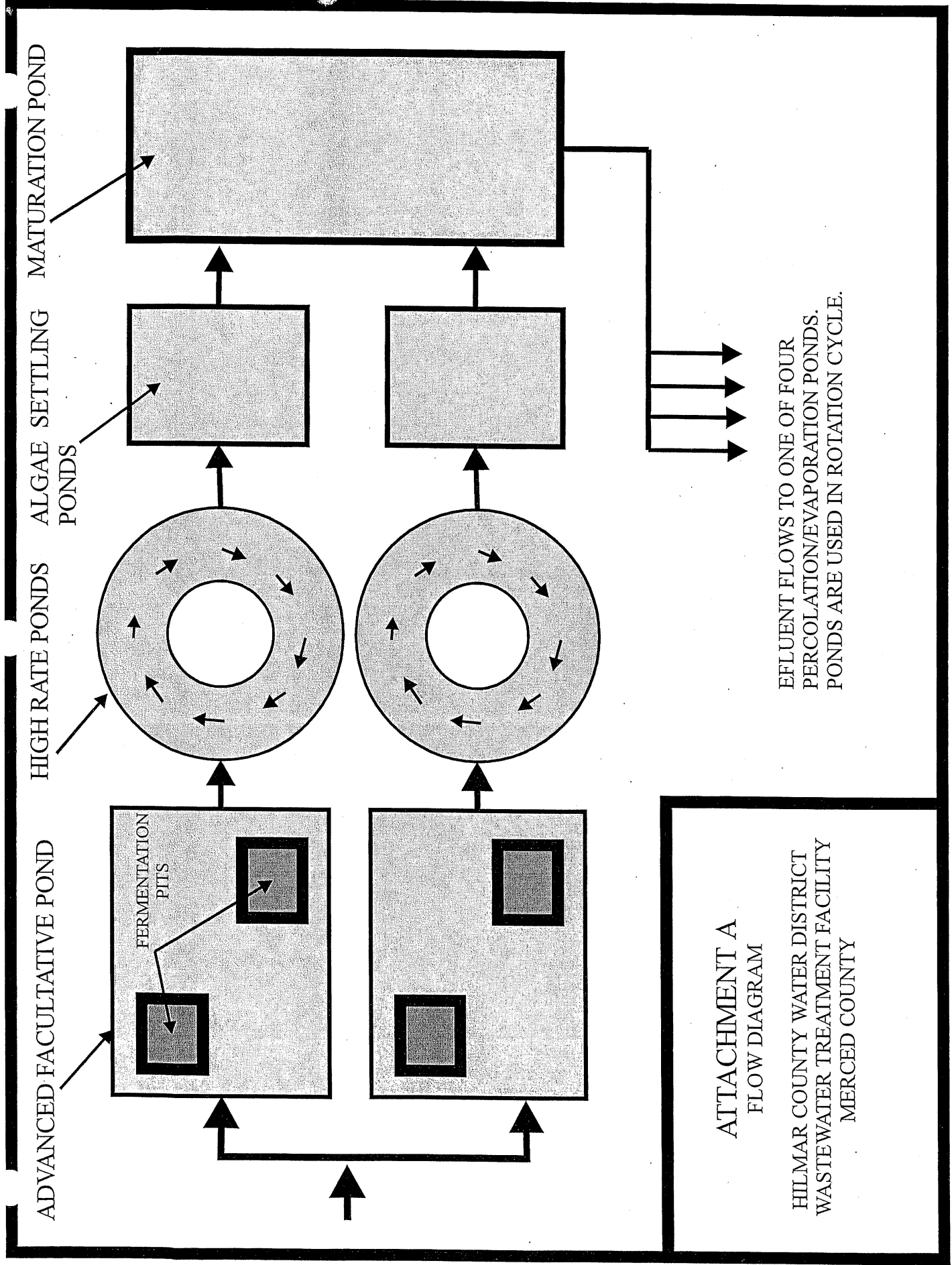
Ordered by:


GARY M. CARLTON, Executive Officer

11 June 1999

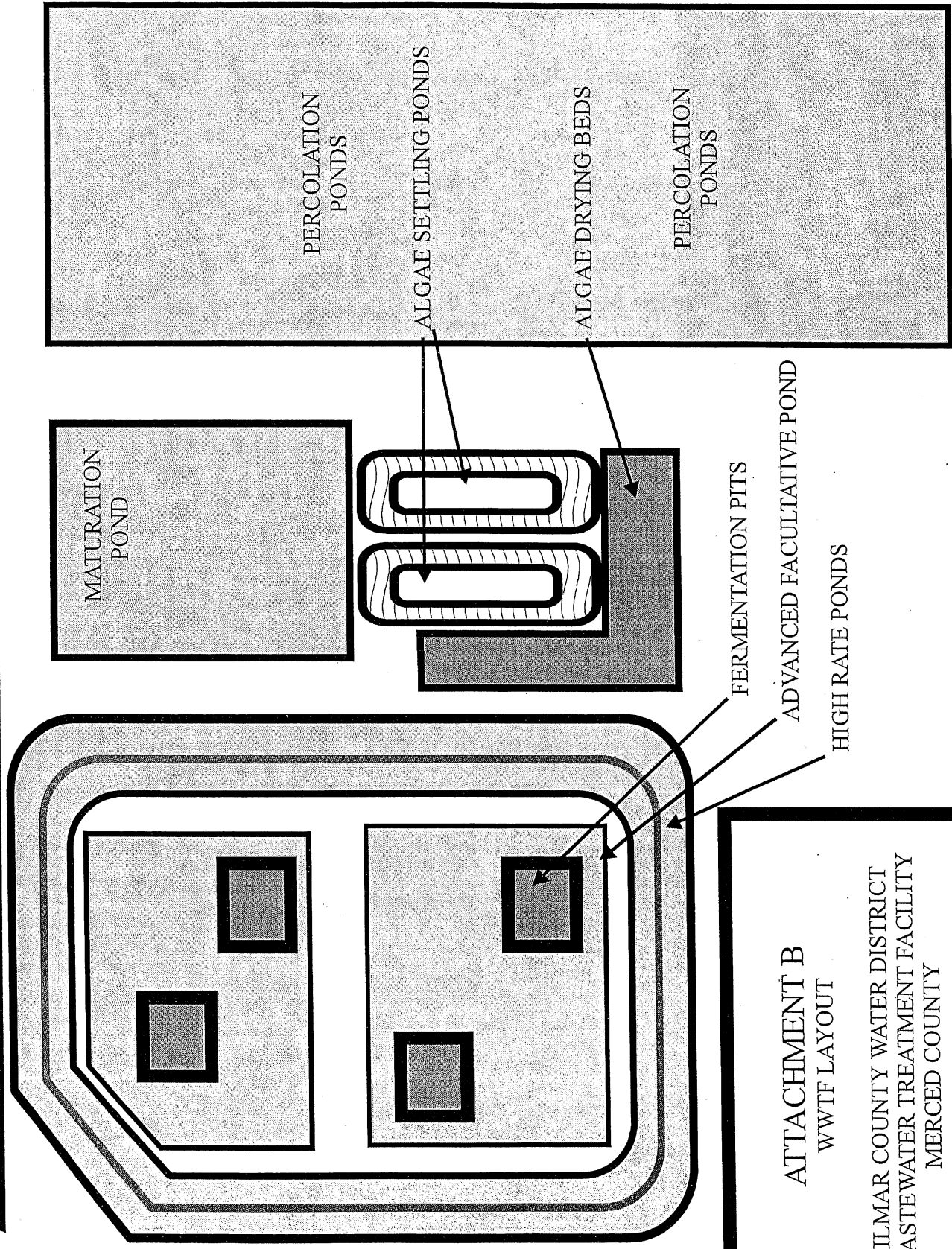
(Date)

CTS:cts/fmc:6/11/99



EFLUENT FLOWS TO ONE OF FOUR PERCOLATION/EVAPORATION PONDS. PONDS ARE USED IN ROTATION CYCLE.

ATTACHMENT A
FLOW DIAGRAM
HILMAR COUNTY WATER DISTRICT
WASTEWATER TREATMENT FACILITY
MERCED COUNTY



MATURATION
POND

PERCOLATION
PONDS

ALGAE SETTLING PONDS

ALGAE DRYING BEDS

PERCOLATION
PONDS

FERMENTATION PITS

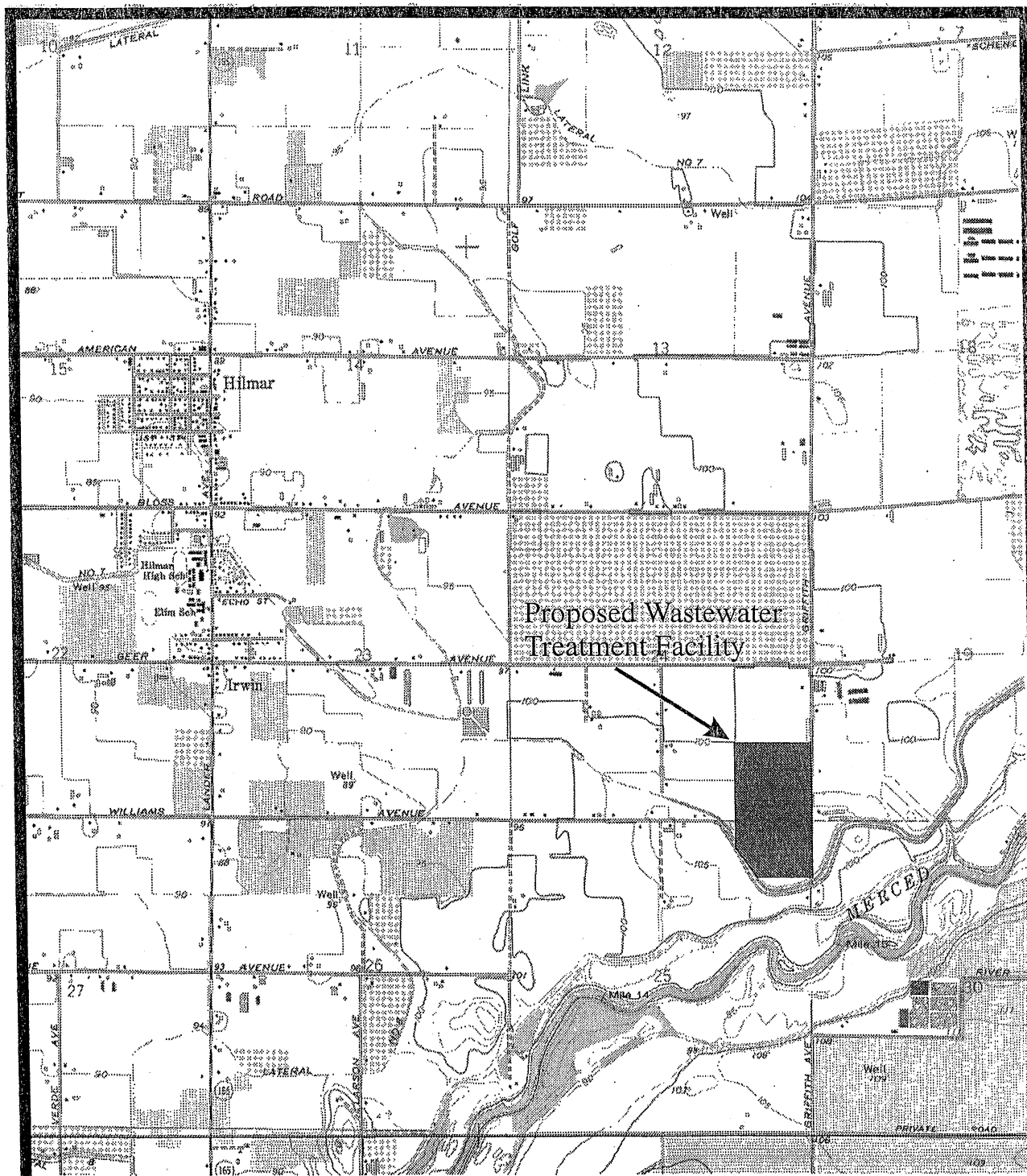
ADVANCED FACULTATIVE POND

HIGH RATE PONDS

ATTACHMENT B

WWTF LAYOUT

HILMAR COUNTY WATER DISTRICT
WASTEWATER TREATMENT FACILITY
MERCED COUNTY



Scale 1" = 1 mile

ATTACHMENT C
VICINITY MAP
HILMAR COUNTY WATER DISTRICT
WASTEWATER TREATMENT FACILITY
MERCED COUNTY

SECTIONS 24 & 25, T6S, R10E, MDB&M
TURLOCK, CA 7.5 Min. U.S.G.S Quad Maps

INFORMATION SHEET

ORDER NO. 99-077
HILMAR COUNTY WATER DISTRICT
WASTEWATER TREATMENT FACILITY
MERCED COUNTY

The Hilmar County Water District (hereafter Discharger) submitted a Report of Waste Discharge (RWD) to replace its existing 0.30 mgd lift station, aeration pond, and two evaporation/percolation ponds (existing WWTF) with a 1.0 mgd Advanced Integrated Wastewater Pond SystemTM having 0.55 mgd disposal capacity (proposed WWTF). Because the existing WWTF is operating at approximately 97% of its design capacity, and is responsible for groundwater degradation, the Board adopted Cease and Desist Order No. 97-103 on 20 June 1997. The proposed WWTF will provide a long term solution to the Discharger's current capacity shortfall and groundwater degradation.

The proposed WWTF will occupy 60-acres about 1/2 mile east of the existing WWTF. It will consist of a lift station, four fermentation pits (FPs), two advanced facultative ponds (AFPs), two high rate ponds, two algae settling ponds, a maturation pond, and four percolation disposal ponds. Floating aerators will be installed in the advanced facultative ponds to provide supplemental aeration during start-up. Effluent from the maturation pond will be discharged to 17.5 acres of proposed evaporation/percolation ponds. The slopes and bottoms of the treatment ponds, with the exception of the maturation pond, will be lined with a synthetic liner to limit percolation losses. After start up of the proposed WWTF, the existing WWTF will be leveled.

The Discharger reviewed opportunities for reclamation and cannot use effluent on irrigated land because it is not economically feasible.

The RWD indicates that the proposed WWTF is primarily underlain by silty sands extending from the surface to depths of about 10 feet below ground surface (bgs). This strata is underlain by interbedded layers of sandy silt, poorly graded sands, and more silty sands to the maximum depth explored of 41.5 feet bgs. The near surface soils are expected to exhibit rapid to moderately rapid permeability. The rate at which the subsurface soils can transmit water is about 0.5 inches per hour.

The RWD indicates that groundwater beneath the proposed WWTF is 16 to 27 feet bgs. General areal groundwater flow is to the south/southeast. Preliminary monitoring well samples indicate groundwater contains total dissolved solids (TDS) concentrations of 140 mg/l to 470 mg/l, nitrate concentrations exceeding 45 mg/l (range of 17 mg/l to 290 mg/l), and total coliform above 23 MPN/100 ml.

The Hilmar County Water District adopted a negative declaration for the proposed project in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines. Mitigation measures to protect against the degradation of groundwater include the placement of monitoring wells around the perimeter of the site to monitor groundwater quality and elevation. While the soil engineer describes a low

INFORMATION SHEET - ORDER NO. 99-077
HILMAR COUNTY WATER DISTRICT ORDER NO.
WASTEWATER TREATMENT FACILITY
MERCED COUNTY

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probability for adversely impacting groundwater, should water level or quality become a problem, the Discharger proposes installation of a subsurface collection drain to intercept water for further treatment or export to an approved site.

This Order is intended to regulate discharge from the replacement WWTF and does not supersede Waste Discharge Requirements Order No. 85-228 for the existing WWTF. Following diversion of all flows to the WWTF regulated by this Order, Order No. 85-228 shall be scheduled for rescission by the Board.

CTS:fmc:6/11/99